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Evaluating the Effectiveness of Contact-Analog and Bounding Box Prototypes in Augmented Reality Head-Up Display Warning for Chinese Novice Drivers Under Various Collision Types and Traffic Density

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ABSTRACT

Augmented Reality Head-Up Display (AR-HUD) is a promising solution to the current warning system distraction problem. However, how to effectively convey warnings through AR graphics is still unclear. This study examines the effectiveness of the contact-analog graphic compared to the bounding box graphic in various collision types and traffic densities. Forty-eight participants watched AR-augmented driving videos and were instructed to respond to critical events. Reaction time, response rate, and subjective evaluations were compared for rear-end and pedestrian collisions in different traffic densities under different warnings. Both bounding box and contact-analog warnings improved driving performance compared to the non-warning group. The contact-analog warning performed better for rear-end collisions, while the bounding box warning had a lower reaction time for pedestrian collisions, regardless of traffic density.

KEYWORDS

AR-HUD; bounding box warnings; contact-analog warnings; world-fixed graphics; conformal graphics; collision type; traffic density

1. Introduction

Driving is a complex task (Ma et al., 2021) that requires drivers to process a large amount of visual information under time pressure (Rusch et al., 2013). However, drivers can only process a small percentage of visual information at once (Eyraud et al., 2015; Ng-Thow-Hing et al., 2013; Rusch et al., 2013; Smith et al., 2021; Ward & Helton, 2022), which can lead to serious road traffic collisions. To address this issue, advanced driver assistance systems (ADAS) have been developed to signal hazards (Brijs et al., 2022). However, many of these systems lack spatial information (Kim et al., 2016; Kim & Gabbard, 2019) and cannot direct drivers' attention to the precise location of the hazard (Kim et al., 2016). Recent research suggests that these hazard warnings may encourage "blind" reactions (Merenda et al., 2018). To overcome this problem, augmented reality-based head-up displays (AR-HUDs) are becoming a promising solution (Tonniss & Klinker, 2006).

Augmented reality (AR) is an interface metaphor that allows for overlaying visual objects onto the physical world, in the form of graphical augmentation (Grubert et al., 2017; Kim & Hwang, 2016; Langlois & Soualmi, 2016; Ma et al., 2021; Ward & Helton, 2022), enabling users to interact with physical and virtual objects simultaneously (Grubert et al., 2017; Kim et al., 2018). In-car AR applications can provide visual information about driving while keeping drivers' eyes on the road (Calvi et al., 2020; Gabbard et al., 2014; Ng-Thow-Hing et al., 2013; Tonniss & Klinker, 2006). AR-HUDs offer the benefit of providing spatiotemporal information on

imminent hazards (Gabbard et al., 2014; Ng-Thow-Hing et al., 2013; Schwarz & Fastenmeier, 2018). Previous studies have shown that AR-HUDs effectively signal hazards (Kim & Hwang, 2016), reduce drivers' inattention blindness (Wang et al., 2021), improve intuitive cognition, and decrease driver distraction (Park et al., 2016).

The design of graphical elements in AR-HUDs greatly affects their effectiveness in delivering warnings (Jing et al., 2022; Kim & Gabbard, 2019). The saliency of the AR graphic impacts attention and driving behavior (Chen et al., 2023; Eyraud et al., 2015; Jing et al., 2022). Poor design can lead to suboptimal performance and reduced trust (Ariansyah et al., 2022; Ng-Thow-Hing et al., 2013). It is crucial to find a balance in designing AR graphics that capture attention without overwhelming or distracting drivers (Eyraud et al., 2015; Park et al., 2016). Thorough investigations are necessary to understand the complexities of AR graphics before the widespread implementation of AR-HUDs in vehicles (Faria et al., 2020).

1.1. Conceptual designs of AR interfaces

Two conceptual designs of AR graphics exist in the context of AR technology: screen-fixed graphics, which are fixed to specific areas within the windshield, and world-fixed graphics (or conformal graphics), which are directly superimposed on real-world objects to blend seamlessly (Gabbard et al., 2014; Kim et al., 2016, 2018; Riegler et al., 2021; Smith et al., 2021). Typically, the screen-fixed graphic can be used

to display important driving information that is not directly related to real objects, such as vehicle speed, fuel level, and engine temperature, while the world-fixed graphics are suitable for presenting information regarding navigation and imminent hazard warnings (Gabbard et al., 2014) as they provide a natural way to convey spatiotemporal information aligned with the moving objects (Ng-Thow-Hing et al., 2013).

There are some AR interface studies focusing on world-fixed graphics (conformal graphics). In general, two major prototypes are involved in the study of world-fixed graphics: namely, virtual shadows (Jing et al., 2022; Kim et al., 2016, 2018) and contact-analog graphics (Faria et al., 2020; Medenica et al., 2011; Schewe & Vollrath, 2021; Smith et al., 2021) (see Figure 1). Virtual shadow is a dome-shaped conformal virtual object, which is commonly used in the field of warning (Kim et al., 2018). Jing et al. (2022) investigated the effect of visual shadows on autonomous vehicle takeover efficiency and drivers' eye movement characteristics. The result demonstrated that the virtual shadow AR-HUD interface could effectively reduce the driver's visual distraction, and improve the efficiency of obtaining visual information. Kim et al. (2016) also used a visual shadow to augment the crossing pedestrian and the evaluation result showed that the visual shadow warning was outperformed in all aspects. Meanwhile, contact-analog graphics, encompassing the visual images within the same general field of view as the critical objects (Wickens & Long, 1994), are commonly used as navigation cues. For example, Medenica et al. (2011) assessed the effectiveness of AR, street view, and traditional maps on personal navigation devices (PNDs). In the case of the AR PND, they used a contact-analog graphic overlaid on the route to provide directional indications. The results revealed that the contact-analog graphic had a positive impact on driving. Additionally, Faria et al. (2020) employed

the contact-analog graphic to enhance navigation landmarks and assessed drivers' glance behavior. The experiment result revealed that the contact-analog graphic condition was linked to prolonged glances. However, the aforementioned studies on the contact-analog graphic focused on navigation, and a dearth of research exists regarding the contact-analog graphic in hazard warnings.

Previous studies suggested that the potential of AR-HUDs to enhance attention toward cued elements while minimizing the impact on attentional resources could be attributed to the conformal nature of the AR graphic (Kim & Gabbard, 2019; Rusch et al., 2013). We hypothesize that contact-analog graphics, similar to visual shadows, have the potential to serve as effective warnings due to their conformal characteristics. According to the proximity compatibility interface-design principle, it is recommended that driving task-related elements be processed simultaneously (Peereboom et al., 2023). Visual shadows are commonly used to integrate hazards and warnings for the purpose of alerting drivers. However, in complex traffic scenarios, visual shadow warnings may inadvertently include other irrelevant objects, thereby requiring drivers to spend extra time to identify the critical hazards. In contrast, contact-analog graphics offer the advantage of connecting the visual warning with the critical object by overlaying color on it without covering other objects. Furthermore, the contact-analog graphics, serving as analogs to the corresponding elements in the distant domain (Chen et al., 2023; Naish, 1964; Rusch et al., 2013), can clearly indicate the type of hazard, even when the hazard is occluded by other objects. However, it is also important to note that the contact-analog graphic completely covers the entire object, which might induce visual illusions due to color-induced masking and overlapping (Schewe & Vollrath, 2021). An excessive presence of color in contact-analog graphics is likely to occlude real-world

(a) Visual shadow graphic



(b) Contact-analog graphic used in navigation



Figure 1. Examples of visual shadow graphics and traditional contact-analog graphics. *Note:* (a) Example of the virtual shadow in action. The virtual shadows move with the movement of the pedestrian. (b) Examples of contact-analog displaying concepts that were used in navigation. Guidance advice: the graphic overlaid on the route to provide directional indications. Road sign visualization: the graphic overlaid on road signs to augment sign information.

information (Merenda et al., 2018). Therefore, it remains to be explored whether contact-analog graphics can be equally applicable to the field of warnings.

Currently, the prevailing approach for delivering warnings in AR-HUDs is through the use of bounding boxes to highlight potential hazards. The bounding box is a technology-driven interface design since it is a common way for machine-vision algorithms to convey discovered dangers (Kim & Gabbard, 2019). It could provide accurate location clues of objects due to their salient geometric characteristic and not occlude the object. Previous studies on AR warnings have predominantly employed bounding boxes to augment representations of pedestrians or vehicles (e.g., Chen et al., 2023; Kim & Gabbard, 2019; Wang et al., 2021; Wu et al., 2022). However, it is worth noting that the bounding box may compromise the ability to predict hazard movements and distract drivers from other critical environmental elements as they may induce attention narrowing (Kim et al., 2016; Kim & Gabbard, 2019). Besides, the bounding box warning can inevitably encompass unrelated objects and may compromise the warning effect.

Additionally, there may be subtle differences in the depth of processing required for the two AR warnings. Previous studies have shown that information processing is affected by variations in graphic shape. For example, processing complex graphics requires more time and cognitive effort than processing simple shapes (Dai et al., 2021). As a result, one of the key distinctions between contact-analog warnings and bounding box warnings is that contact-analog warnings outline the contour of the impending hazard and include more complex shape presentations. This might necessitate the driver to allocate extra time to identify and connect the type of hazard represented by the contour. Limited research has investigated the effects of graphics in the context of differences in the level of perceptual processing of graphics. The potential impact of this processing difference justifies further exploration. Therefore, it is necessary to directly compare the effectiveness of the contact-analog and bounding box prototypes in warning scenarios.

1.2. Environment factors and AR graphics

AR-HUD warnings can be applied to two primary driving collision scenarios: rear-end collisions and pedestrian collisions. The rear-end collision is one of the most frequent types of crashes (Muhrrer & Vollrath, 2011; Yue et al., 2021). It accounted for 29% of all crashes in the United States (Yue et al., 2021). Furthermore, vulnerable road users, such as pedestrians, are more likely to be involved in crashes and experience more severe consequences (Schubert et al., 2023; Zhu et al., 2023). Several previous studies have examined the impact of AR warnings in various scenarios, including pedestrian crossings (e.g., Calvi et al., 2020; Rusch et al., 2013), crossing vehicles (e.g., Schwarz & Fastenmeier, 2018), emergency U-turns by the vehicle in front, and leading vehicles braking (e.g., Jing et al., 2022). However, it is important to note that these studies often did not analyze

the effects of AR warnings in these specific scenarios in depth, nor did they consider the potential influence of scenario characteristics on the effectiveness of the AR graphic warning.

Moreover, it is crucial to consider various traffic-density conditions when designing AR-HUD warning interfaces. Previous research indicates that increasing the number of items in the central vision results in extended reaction times and reduced accuracy in detection tasks (Lee et al., 2023). This implies that a complex environment can induce visual clutter, thereby impacting object detection (Li et al., 2019; Rosenholtz et al., 2007). In driving scenarios characterized by high traffic density, which can be perceived as a form of dynamic “situational clutter” (Edquist & Johnston, 2008; Trick et al., 2010), selective attention may be compromised (Edquist & Johnston, 2008), rendering it more challenging to perceive hazards and warnings (Trick et al., 2010). We hypothesize that specific AR warning graphics may be sufficiently noticeable in low-traffic-density conditions but may exhibit reduced effectiveness in high-traffic-density conditions. Additionally, the attention-grabbing nature of AR graphics (Chen et al., 2023; Kim & Gabbard, 2019) could contribute to additional visual clutter, especially caused by object occlusion (Beck et al., 2010). The question of whether providing additional AR warnings in high-traffic-density conditions would enhance hazard detection or exacerbate visual clutter remains unresolved. It is imperative to conduct further investigation into these graphical effects across varying traffic-density conditions.

Consequently, this study aims to investigate the effects of AR warnings on rear-end collisions and pedestrian collisions under varying traffic densities, to further examine their applicability in diverse driving scenarios.

1.3. The objective of the study

Despite the potential of contact-analog graphics in hazard warning, there is a lack of extensive research on their effectiveness in this context, and few studies have directly compared them to bounding box graphics. Furthermore, few research has systematically explored the potential impact of differences in cognitive processing levels on the effectiveness of warnings for the two types of graphics. Therefore, the objective of this study was to examine the strengths and limitations of these two prototypes of AR warnings, specifically considering collision type and traffic density as factors within driving scenarios.

This study also focuses on the effects of AR warnings on novice drivers, who are generally more prone to collisions (Pitt et al., 2021). Previous research suggests that novice drivers tend to devote excessive attention to vehicle handling routines, resulting in error-prone search patterns (Twisk, 1995). Consequently, they may struggle to identify road hazards in complex driving environments. AR technology offers a promising solution by providing visual cues that guide the attention of novice drivers toward the most critical hazards (Rane et al., 2016). However, it is important to consider that novice drivers lack experience, which may potentially affect

their perception of visual warning graphics (Zhao et al., 2023). Thus, there is a need to investigate which AR heads-up display (HUD) interfaces can effectively enhance the risk perception of novice drivers.

Furthermore, previous experiments have mainly been conducted using driving simulators without a genuine AR-HUD device, either relying on flat projections to display videos (e.g., Eyraud et al., 2015; Hwang et al., 2016; Hwang & Kim, 2016; Kim & Gabbard, 2019; Ng-Thow-Hing et al., 2013) or using virtual reality (VR) (e.g., Abdi et al., 2017; Ma et al., 2021; Merenda et al., 2019). It should be noted that, when using a genuine AR-HUD device, the AR graphic is projected directly onto the windshield, causing our eyes to converge on points beyond the windshield, which can result in a distracting double image of the AR graphic (Ng-Thow-Hing et al., 2013). Therefore, the varying distances between the AR graphic and the real object can cause a conflict between vergence and accommodation, impacting depth perception (Adams et al., 2022; Gabbard et al., 2014; Navarro et al., 2017; Ng-Thow-Hing et al., 2013; Werneke & Vollrath, 2013; Winkler et al., 2016). Merely relying on flat projection or VR could introduce a confounding factor in assessing the influence of the AR graphic.

Above all, the main objective of this study is to assess the advantages and drawbacks of various AR warnings for novice drivers in a genuine AR-HUD context across different driving scenarios.

2. Methods

2.1. Participants

Due to limitations in G*power's capability to handle a three-factor mixed design, a priori power analysis was performed using MorePower 6 (Campbell & Thompson, 2012) with a significance level of 0.05, power of 0.8, and effect size of 0.2. The results indicated that the total sample size should exceed 42 to achieve the desired power. The study recruited a convenience sample of 48 gender-balanced participants from Zhejiang Sci-Tech University in Hangzhou, China. Participants ages ranged from 19 to 26 years, with a mean age of 22 years ($M = 22$, $SD = 2.0$). Additionally, all participants held a valid Chinese driver's license, with an average driving experience of 1526 km and 2.7 years. The study focused on novice drivers, and none of the participants had any prior experience with the AR-HUD. All participants confirmed having a normal or corrected-to-normal vision, provided written informed consent in accordance with the approved research protocol of the Institutional Review Board at the Department of Psychology, Zhejiang Sci-Tech University, and received compensation for their participation.

2.2. Experimental design

A three-factor mixed design was employed to investigate the impact of AR warnings on driving performance and usability. The design included AR warning type (bounding box warnings vs. contact-analog warnings vs. non-warning) as a

between-subject variable, traffic density (high traffic density vs. low traffic density), and collision types (rear-end collision vs. pedestrian collision) as within-subject variables.

We adopted the definition of traffic density provided by Trick et al. (2010), where high traffic density was defined as one vehicle per 150 m, and low traffic density was defined as one vehicle per 1500 m.

Dependent variables for comprehensive assessment included behavior performance indicators, such as reaction time and response rate. Additionally, to evaluate user experience related to the two tested AR graphics, participants completed the User Experience Questionnaire (UEQ) (Laugwitz et al., 2008) after the experiment.

2.3. Experimental equipment

The experimental equipment setup closely resembles the one used in the study by Hwang et al. (2016). The experiment involved several equipment components: a 180-inch screen, a beam projector (CB-935W, Epson, JPN) with 3700 luminance, a keyboard, a computer synchronizer, and two computers. One computer was used for collecting driver response data, while the other computer projected the background image. In the optical see-through augmented reality display condition, the AR-HUD device (DGG01, Zhangjiagang Qidian Photoelectric Technology Co., Ltd., CN) was used to play the videos. The AR-HUD is developed with dual free-form reflective optics. Its corresponding Digital Micromirror Device (DMD) resolution is 864×480 , the brightness is 12,000 nit, and the refresh rate is 60 Hz. Presenting the physical background at a closer distance beyond AR virtual images can induce the convergence adaptation conflict (Huang & Menozzi, 2022). We placed an LED screen projecting a picture of mountains at a vertical distance of ~ 3 m from the viewers to imitate the convergence adaptation conflict in AR application. The experiment scenarios were built using a driving simulator, which included driving simulator software (STISIMDRIVE M100K, Systems Technology, Inc., Hawthorne, CA, USA), and an operating system (Logitech MOMO, Newark, CA, USA). The experimental setup is illustrated in Figure 2.

2.4. Materials

Due to technical limitations, it was not possible to directly augment target objects in the driving simulator. Therefore, driving videos generated by a medium-fidelity driving simulator were used in the study.

The experiment consisted of two types of traffic conditions: high traffic density and low traffic density. There was a total of 24 videos, with each experimental video having a playback time of ~ 1.5 min. Both traffic conditions consisted of the same types of latent threats, such as rear-end collisions with preceding vehicles and collisions with pedestrians. Previous research indicated that drivers had longer reaction times to pedestrians appearing on the left side compared to the right side (Powelleit & Vollrath, 2019). The direction of

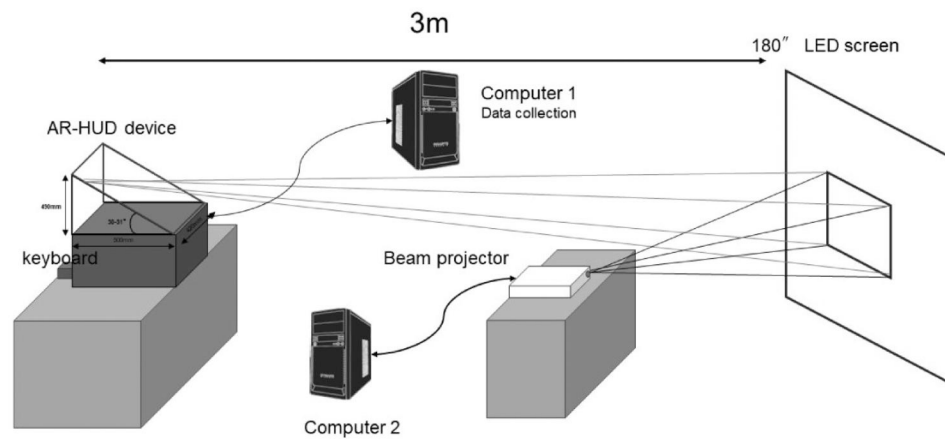


Figure 2. Demonstration graphs for experiment laboratory.

pedestrian crossings was balanced, and the order of left and right crossings was randomized to reduce potential bias.

Furthermore, color can significantly contribute to the perceived meaning and visibility of graphics (Kim et al., 2016) and previous research demonstrated that color influences hazard perception (Ng & Chan, 2018). Utilizing appropriate color coding could enhance performance (Ng & Chan, 2018). In traditional user interface design (UID), caution indications are typically represented in yellow, while warning indications are represented in red (Friedrich & Vollrath, 2022). However, the presentation of red and yellow is affected by hue shifts in AR projections. For example, some argued that the color yellow is more prominent in AR (Merenda et al., 2016), while the color red is less prominent in AR (Gabbard et al., 2005). Therefore, to avoid the interference of hue shift in AR, each participant was shown red and yellow versions of the AR warning. Both colors were in RGB space. Red was [255, 0, 0], and yellow was [255, 255, 0].

The experiment divided the 24 videos into four blocks based on the combination of traffic density and warning color: high traffic density with a red warning, high traffic density with a yellow warning, low traffic density with a red warning, and low traffic density with a yellow warning. Each video served as a trial, with six trials in each block. The presentation order of the four blocks was counterbalanced using Latin squares. Since the videos were prerecorded, participants were unable to control the vehicle. To closely simulate real driving conditions, the videos were interspersed with other unpredictable events, such as speeding, cutoff vehicles, and red-light violations. Each group of videos included 15 events, including two pedestrian collisions, one rear-end collision, cutoff vehicles, speeding, and red-light violations.

Two types of AR warnings were employed, namely, the bounding box warning and the contact-analog warning, to alert drivers about unforeseen events. The contact-analog warning used in this study was integrated into the virtual environment, where its shape dynamically changed in response to the movement of the augmented object (Smith et al., 2021), whereas the shape of the bounding box warning remained unchanged. Providing a warning with a duration

of 3.0–4.0 s before a possible collision may be appropriate (Drüke et al., 2018; Ma et al., 2021; Zhang et al., 2015) and may significantly reduce driving speed (Hajiseyedjavadi et al., 2018). In this experiment, a pre-recorded video was utilized, maintaining a constant speed of 54 km/h for the ego-vehicle. Taking into account the driving speed in the video, we pre-determined the parameters (such as distance and speed) of the leading vehicle and pedestrian in each scenario. This ensured that the AR warning, lasting for 3.0 s, could be presented until the collision occurred in the various collision scenarios. Figures 3 and 4 depict AR warnings in augmented conditions.

2.5. Procedure and tasks

Informed consent was obtained from the participants before their first visit. Subsequently, participants completed a demographics questionnaire. They were then comfortably seated in front of the AR-HUD. Participants were instructed to look through the device and direct their gaze toward the virtual panel in front of them. The experimenter familiarized the participants with the AR warning.

The participants were divided into three groups for the experiment: a contact-analog warning group, a bounding box warning group, and a control group without any warnings serving as the baseline. All participants in the three groups had the same physical settings except that the presentation of the warning differed. Although we did not provide the control group with the AR warning, the participants in this group still watched the video through the AR-HUD to avoid possible interference caused by the difference in physical settings.

Participants were instructed to imagine themselves as the driver while watching the video. In the videos, the ego vehicle is traveling on the road at a constant speed of 54 km/h. In each group of videos, the vehicle passes through road sections, such as intersections, curves, uphill grades, and straight roads. We added these road sections to increase the realism of the scene as much as possible. During the experiment, we asked the participant to press a button promptly upon observing unpredictable events, including

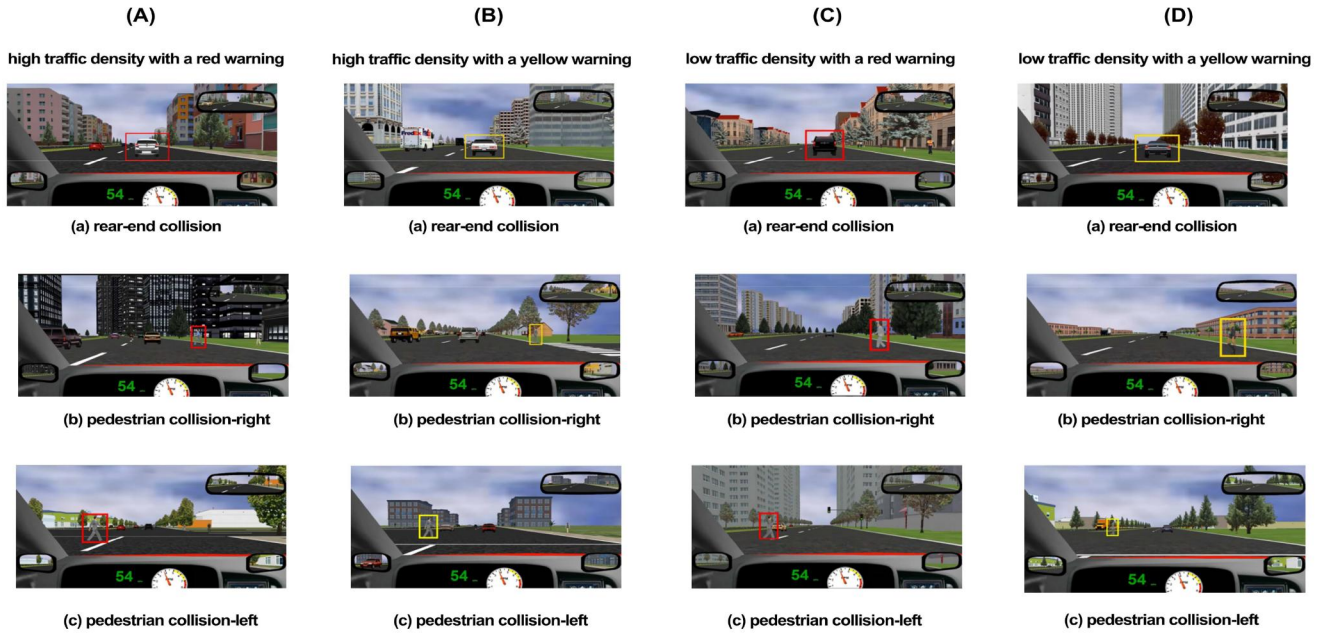


Figure 3. The bounding box warning examined in the driving simulator study. *Note:* We divided the videos into four groups based on traffic density and warning colors. The videos in group (A) have a high traffic density, with the red warning. The videos in group (B) have a high traffic density, with the yellow warning. The videos in group (C) have a low traffic density, with the red warning. The videos in group (D) have a low traffic density, with the yellow warning. Each group of videos includes three types of critical events: (a) rear-end collision, (b) right-side pedestrian collision, and (c) left-side pedestrian collision, and left and right-side pedestrian collisions occur equally.

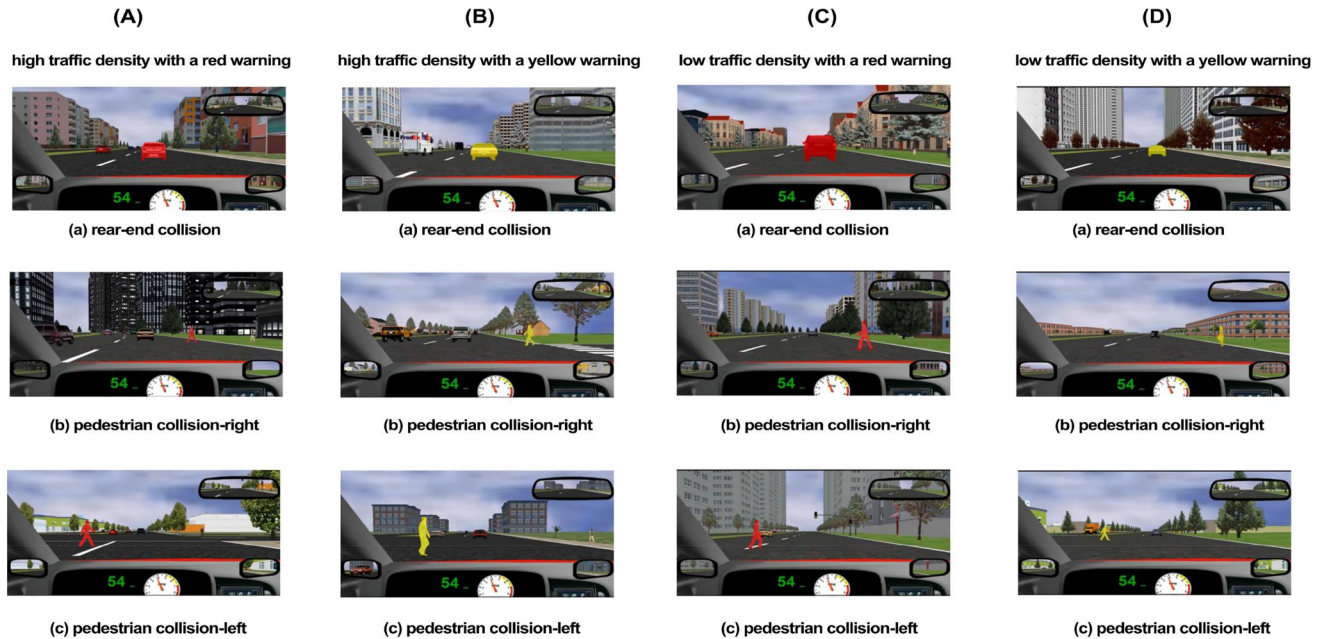


Figure 4. The contact-analog warning examined in the driving simulator study. *Note:* The videos were categorized as Figure 2.

vehicle speeding, vehicle running a red light, vehicle lane changing, preceding vehicle braking (rear-end collisions), and a pedestrian crossing the road in front of the ego vehicle (pedestrian collisions). Participants were required to respond to all events to prevent anticipation of pedestrian crossings and vehicle braking (Wang et al., 2021). Additionally, the first three events were designed to enhance participants' focus on the experiment. Upon completion of all blocks, participants filled out a User Experience Questionnaire (UEQ). Figure 5 illustrates the detailed procedure.

3. Results

The experimental data were analyzed using IBM SPSS Statistics 26. A significance level of 0.05 was used for all statistical analyses. Generalized estimating equations (GEE), based on the work of Nelder and Wedderburn (1972) on generalized linear models, were used for the statistical analyses. GEE is a versatile statistical approach used for analyzing longitudinal and clustered data by fitting a marginal model. Given the repeated measures nature of this experiment data,

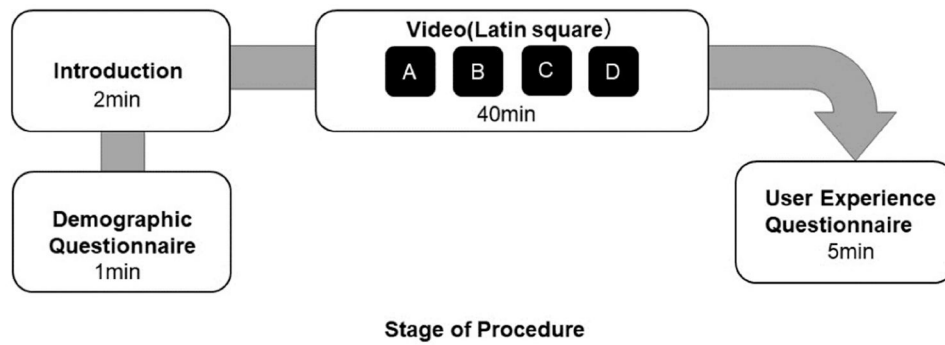


Figure 5. Stage of procedure.

there may be a correlation between the data. Liang and Zeger (1986) in their study suggested with repeated observations, the correlation among values for a given participant must be taken into account, and the estimating equation might be suitable to analyze the data. In addition, GEE is well-suited for situations involving missing data and small sample sizes. Due to participants not responding to certain unpredictable events, the behavioral performance data collected in this study contained missing values. GEE was employed to account for these missing data points. Last but not least, we examined the distribution of the reaction time and the response rate and found that the data in this experiment were skewed. GEE relaxes the data distribution requirements and thus is more suitable for analyzing the data in this case.

In sum, behavioral performance data were analyzed using GEE, with AR warning types, traffic density, and collision types serving as the independent variables. The effects of each predictor and their interactions were evaluated by examining the model effects, parameter estimates, and estimated marginal means. Estimated marginal means were utilized to compare the effects of AR warnings across different conditions, with the Bonferroni correction applied in the GEE *post-hoc* test.

3.1. Reaction time

The collision type had a significant main effect on reaction time [Wald $\chi^2(1) = 124.228$, $p < 0.001$]. The reaction time was significantly shorter in rear-end collisions compared to pedestrian collisions. The AR warning type had a significant main effect [Wald $\chi^2(2) = 67.034$, $p < 0.001$], whereas traffic density [Wald $\chi^2(1) = 0.073$, $p = 0.787$] did not have a significant effect. The results indicated that both the AR warnings could effectively decrease the reaction time but the contact-analog warning was linked to a longer reaction time compared to the bounding box warning ($p = 0.012$).

There was a significant interaction between AR warning types and collision types [Wald $\chi^2(5) = 391.539$, $p < 0.001$]. In the pedestrian collision condition, GEE *post-hoc* tests revealed that both AR warning groups were accompanied by shorter reaction times when compared to non-warning groups (the contact-analog warning: $p < 0.001$; the bounding box warning: $p < 0.001$), and the reaction time was longer under the contact-analog warning compared to the bounding box warning

($p < 0.001$). Moreover, in the rear-end collision condition, GEE *post-hoc* tests revealed that both AR warning groups could effectively reduce the reaction time compared to the non-warning group (the contact-analog warning: $p < 0.001$; the bounding box warning: $p < 0.001$), but no significant difference between the bounding box warning and the contact-analog warning ($p = 0.293$). These results suggested that the different impact of the AR warning was primarily evident in the perception of pedestrians.

There was a significant interaction between collision types and traffic density [Wald $\chi^2(3) = 156.085$, $p < 0.001$]. In the rear-end collision condition, GEE *post-hoc* tests indicated that reaction time was longer under the low traffic density compared to the high traffic density ($p = 0.011$), whereas no significant difference was found between the traffic density in the pedestrian collision condition ($p = 0.055$).

There was a significant interaction observed between traffic density and AR warning types [Wald $\chi^2(5) = 91.171$, $p < 0.001$]. GEE *post-hoc* tests revealed that both AR warning groups were significantly better than the non-warning group in the high-traffic density condition (the contact-analog warning: $p < 0.001$; the bounding box warning: $p < 0.001$) and in the low-traffic density condition (the contact-analog warning: $p < 0.001$; the bounding box warning: $p < 0.001$). However, it is worth noting that the bounding box warning was related to a shorter reaction time than the contact-analog warning (in the high traffic density condition: $p = 0.019$; in the low traffic density condition: $p = 0.015$).

Significant interactions were observed for the three-way interaction effect. Our findings revealed a significant interaction between collision types, traffic density, and AR warning types [Wald $\chi^2(11) = 445.855$, $p < 0.001$]. The effect of the AR warning groups was better than the non-warning group in different situations. However, the effect of the two AR warning groups varied in different conditions. In the pedestrian collision condition, the reaction time was lower when using the bounding box warning compared to the contact-analog warning for both high ($p < 0.001$) and low ($p = 0.003$) traffic densities. However, in the rear-end collision condition, no significant difference between the bounding box warning and the contact-analog warning in both high ($p = 0.613$) and low ($p = 0.124$) traffic density conditions. The results indicated that it was appropriate to consider the use of different warning graphics depending on

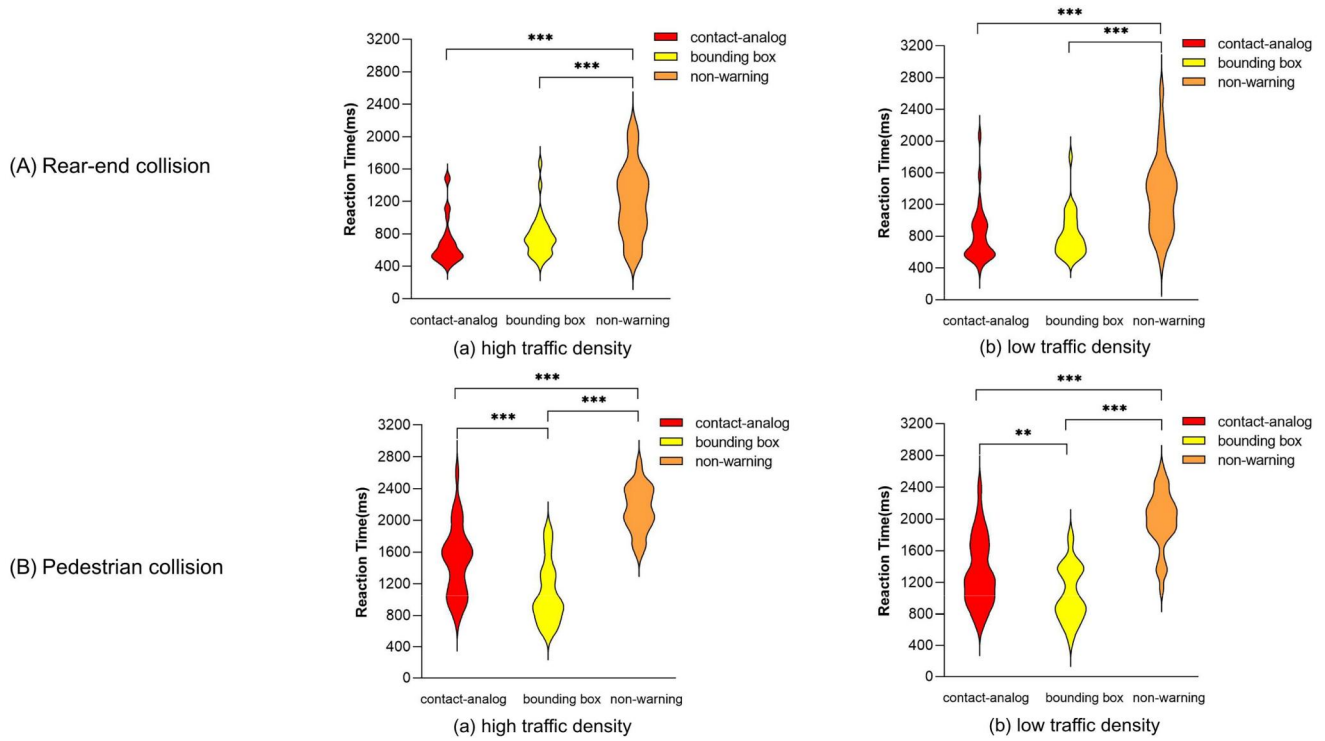


Figure 6. Effect of AR warning types and traffic density on reaction time under rear-end collision (A) and pedestrian collision (B).

Table 1. Corresponding mean (SD) of reaction time and response rate of AR warnings under different driving conditions.

Dependent variable	AR warning type	Rear-end collisions		Pedestrian collisions	
		High traffic density	Low traffic density	High traffic density	Low traffic density
Reaction time	CA	686.66 (284.27)	804.28 (350.58)	1507.70 (435.57)	1354.52 (433.88)
	BB	835.28 (426.96)	824.51 (285.61)	1062.83 (392.85)	1051.32 (347.90)
	Non	1188.09 (456.07)	1341.51 (501.15)	2131.44 (296.16)	1976.30 (370.91)
Response rate	CA	96.87% (0.176)	96.88% (0.176)	87.50% (0.336)	96.87% (0.176)
	BB	96.87% (0.176)	87.50% (0.336)	93.75% (0.245)	96.87% (0.176)
	Non	96.88% (0.176)	93.75% (0.245)	81.25% (0.396)	90.62% (0.296)

CA: the contact-analog warning; BB: the bounding box warning; Non: non-warning.

the collision types. The detailed results can be found in Figure 6. Table 1 summarizes the descriptive result under different conditions.

3.2. Response rate

As for response rate, there was no significant effects were observed for collision types [Wald χ^2 (1) = 1.345, p = 0.246], traffic density [Wald χ^2 (1) = 0.024, p = 0.876], and AR warning types [Wald χ^2 (2) = 0.733 p = 0.693].

Similarly, no significant interactions were found in the results for the response rate. That is, collision type and traffic density [Wald χ^2 (3) = 5.680, p = 0.128], collision type and AR warnings [Wald χ^2 (5) = 4.713, p = 0.452], traffic density and AR warning type [Wald χ^2 (5) = 0.921, p = 0.969], and collision type, traffic density, and AR warning type [Wald χ^2 (11) = 9.159, p = 0.607] did not have an interaction. We suspected that this result might be related to the fact that unexpected hazards (i.e., leading vehicles and crossing pedestrians) end up very close to directly in front of the driver, so drivers were able to respond to unexpected hazards in most cases. Therefore, the response rate did not reach significant differences among the three groups in

Table 2. GEE results for reaction time and response rate under different conditions.

	Reaction time		Response rate	
	Wald χ^2	p	Wald χ^2	p
Main effects				
Collision type	124.228	<0.001***	1.345	0.246
Traffic density	.0073	0.787	.024	0.876
AR warning type	67.034	<0.001***	.733	0.693
Two-factor interactions				
Collision type*Traffic density	156.085	<0.001***	5.680	0.128
Collision type*AR warning type	391.539	<0.001***	4.713	0.452
Traffic density*AR warning type	91.171	<0.001***	.921	0.969
Three-factor interactions				
Collision type*Traffic density*AR warning type	445.855	<0.001***	9.159	0.607

Note. *** p < 0.001.

different situations. We further investigated the above results by examining the GEE *post-hoc* results. We found that while not reaching a significant difference, the response rate of the non-warning group was overall lower than that of the warning groups. Moreover, the bounding box warning group had a lower response rate than the contact-analog warning group in rear-end collisions, this suggested that the contact-analog warning group might have a slight advantage in rear-end collisions. Table 2 presents the GEE result of AR warning

Table 3. The mean score, standard deviation (SD), and the comparative result of UEQ for AR warnings.

UEQ	Contact-analog warning		Bounding box warning		<i>p</i>
	Mean	SD	Mean	SD	
Attractiveness	1.15	1.04	1.66	0.86	0.14
Perspicuity	1.08	1.26	2.25	0.52	0.003**
Efficiency	1.27	0.99	1.67	0.85	0.22
Dependability	1.13	0.89	1.78	0.68	0.026*
Stimulation	0.63	0.89	0.92	1.02	0.39
Novelty	0.94	1.03	0.75	1.38	0.67

Note. * $p < 0.05$, ** $p < 0.01$.

types, collision types, and traffic density on the reaction time and response rate.

3.3. UEQ score

The UEQ provides scores on six different scales: attractiveness, perspicuity, efficiency, dependability, novelty, and stimulation - each of them was examined separately. Additionally, five benchmark levels (excellent, good, above average, below average, and awful) can be utilized to further categorize scores, facilitating comparisons across a broader range of UEQ ratings (Plewan et al., 2021; Schrepp et al., 2017).

The six subscales of the UEQ offer a more comprehensive evaluation of the two AR warning types. The bounding box warning received significantly higher ratings on the perspicuity scale ($p = 0.003$) and the dependability scale ($p = 0.026$). For perspicuity and dependability scales, the bounding box warning was rated as excellent, while the contact-analog warning was rated as below average for both. Although there were no statistically significant differences on the scales of attractiveness ($p = 0.14$), efficiency ($p = 0.22$), and stimulation ($p = 0.39$), the bounding box warning was still correlated with higher scores. Additionally, it is noteworthy that the bounding box warning received a lower score than the contact-analog warning in terms of novelty ($p = 0.67$). Overall, the bounding box warning was rated as below average in stimulation only, whereas the contact-analog warning received below-average ratings in the categories of attractiveness, perspicuity, dependability, and stimulation. Table 3 presents the detailed results of the UEQ scores.

4. Discussion

The results of this study demonstrate that the two AR warnings have varying effects in different collision types. For instance, in the pedestrian collision scenario, the bounding box warning proves to be more effective, whereas, in the rear-end collision scenario, the contact-analog warning appears to hold a slight advantage. In the subsequent section, we will initially analyze the effects of the two AR warnings in different collision types to summarize the value of their application in the design of future AR-HUD warning interfaces. Subsequently, considering the disparities in the level of perceptual processing between the two AR warnings, we will further investigate the impact of the level of graphical processing on the warning from a cognitive perspective. We aim for the analysis from a cognitive processing

perspective to broaden future research directions in the field of warning.

4.1. Consideration of AR warnings on AR-HUD warning interface design

The findings reveal that all AR warnings exhibited a significant enhancement in task performance in terms of reaction time when compared to the non-warning group, for both high and low-traffic density conditions. Notably, the bounding box warning exhibited a significantly faster reaction time in comparison to the contact-analog warning. This result partially confirmed that both AR warning types were effective in the AR-HUD warning interface, but the contact-analog warning might not be as effective as we expected.

Considering the collision types, a distinct disparity in the impact of the two AR warnings was observed. The driver's reaction time did not significantly differ between the contact-analog warning and the bounding box warning in rear-end collisions. However, in pedestrian collisions, the bounding box warning considerably reduced the driver's reaction time compared to the contact-analog warning. This result aligns with a prior study by Jing et al. (2022). They compared two AR warning interfaces in various scenarios and the result similarly demonstrated that AR warnings showed different effects in different scenarios. Evidently, scenarios could affect hazard detection (Crundall et al., 2012), and lead drivers to adapt their reactions accordingly (Winkler et al., 2018). As such, the scenario characteristics can influence the effectiveness of AR warnings (Hwang et al., 2016). This scenario effect could potentially be attributed to the disparate positions of these two scenarios within the driver's field of view.

In the course of driving, the leading vehicle consistently remains within the driver's central vision (longitudinal incidents), whereas unpredictable pedestrians frequently enter the driver's peripheral vision (lateral incidents). Extensive research has substantiated that the fidelity of representation varies across the visual field (Rosenholtz, 2011). Although information in the initial forward field of vision was accurately reported, details in the peripheral field were either disregarded or not registered (Rosenholtz, 2011; Thomas & Wickens, 2001). Additionally, Dozza (2013) highlighted significantly longer response times exhibited by drivers during lateral incidents compared to longitudinal incidents. Consequently, the driver excels in processing information within the central vision but displays relative inefficiency in processing information within the peripheral vision (Chen et al., 2023).

Hence, in the case of rear-end collisions, the provision of a warning, be it a contact-analog or bounding box warning, proved adequate in drawing the driver's attention and enhancing their response. However, the findings also demonstrated a greater response rate of the contact-analog warnings (albeit statistically insignificant) in rear-end collisions. One potential explanation could be associated with its precise overlay on the imminent hazards through color. Previous studies involving neurological patients have

demonstrated that color discrimination can occur without conscious awareness (Breitmeyer et al., 2004; Tao et al., 2017; Tarenskeen et al., 2015; Ward & Helton, 2022). And the size of color patches has an impact on our perception (Badiane et al., 2017; Johnson, 2014). Contact-analog warnings were created by using a large color-covered area on the leading car, which had a more significant presence in the central vision than the bounding box warning. The blending of colors between the AR warning and the physical object can lead to enhanced visual connectivity, which positively impacts the driver's perception (Jing et al., 2022). Thus, the contact-analog warning exhibited a higher level of effectiveness compared to the bounding box warning, especially within the central vision, resulting in a higher response rate.

However, in the case of pedestrian collisions, the impact of AR warning types displayed a noteworthy disparity. The results indicated that drivers exhibited longer response times under contact-analog warning while the bounding box warning offered an effective cautionary measure for pedestrian crossings, ensuring optimal human performance in AR driving conditions. It signified that color-based searching was less effective in the pedestrian collision condition. Interestingly, prior studies suggested that color-based searching was more efficient than shape-based searching (Hwang et al., 2016; Ng-Thow-Hing et al., 2013). This conflict discovery prompted us to re-evaluate the distinctive attributes of warnings that could impact perception during the search process, while also factoring in the statistically significant variations in effects between the bounding box warning and the contact-analog warning in pedestrian collisions.

The differential saliency of the contact-analog and bounding box warnings in peripheral vision offers one possible explanation for the contradictory findings. The contact-analog warning was incorporated into the actual environment by utilizing the pedestrian's geolocation, causing its apparent size (as measured by relative size and height in the driver's visual field) to adjust as the driver approached the pedestrian in question (Kim et al., 2018). Typically, pedestrians initially come into view in the peripheral vision from a distance, and then gradually move toward the driver's central vision (Domeyer et al., 2022). As a result, when pedestrians appeared on the peripheral roadside, the contact-analog warning occupied a relatively small area in the driver's field of vision, thereby reducing its saliency. Additionally, the color-induced masking and overlapping within the contact-analog warning may create the perception of an object located further away, potentially leading to erroneous driver decisions (Ng-Thow-Hing et al., 2013). Furthermore, the presence of peripheral pedestrians did not elicit the same level of stimulation in drivers compared to the preceding vehicles in the center (Hwang et al., 2016). The peripheral vision generally presents higher thresholds for most perceptual tasks compared to the central vision (Finlay, 1982). Thus, employing color as an exogenous cue to enhance peripheral pedestrian detection was effective only within the active field of view (Ward & Helton, 2022). Taking all these factors into consideration, drivers exhibit slower response times to the contact-analog warning. Conversely, while the bounding box warning

may have covered extraneous objects, it covers a larger area compared to the contact-analog warning and is more salient in the peripheral field of view to some extent. This suggests that the positive effect of a larger area of the bounding box warning may offset the negative effect of covering other objects in the peripheral field of view. Earlier studies have also shown that the presence of a bounding box substantially enhanced driver performance in response to unpredictable peripheral events (e.g., Chen et al., 2023; Wang et al., 2021). Above all, in the peripheral field of view, a bounding box warning may be more effective.

The subjective attitude toward the different AR warnings was assessed by obtaining ratings from the user experience questionnaire (UEQ). This subjective data serves as a valuable complement to behavioral data (Plewan et al., 2021). The results of the subjective rating aligned with the behavioral data. The bounding box warning was generally rated higher in terms of attractiveness, perspicuity, efficiency, dependability, and stimulation. Particularly, the bounding box warning was rated as excellent in perspicuity and dependability scales, while the contact-analog warning was rated as below average for both. However, it is worth noting that drivers perceived the contact-analog warning as more novel, although the differences were not statistically significant.

Overall, our findings indicate that future AR-HUD warning interfaces need to consider the specific characteristics of different scenarios and use appropriate AR warnings accordingly. In pedestrian collision scenarios, we advocate for the use of bounding box warnings to improve the detection of peripheral pedestrians. Such warnings are more likely to attract attention in the peripheral vision, thus effectively alerting the driver. Conversely, in rear-end collision scenarios, we propose employing contact-analog warnings to specifically alert the driver of the leading vehicle's braking actions. These warnings can effectively communicate a sense of danger by displaying a significant color area in the driver's central vision.

4.2. Investigation of the impact of graphics processing levels on warning

Additionally, the perspective of perception provides an additional framework for interpreting our findings. Previous studies have suggested that understanding human perception of graphics can inspire shape analysis, pattern recognition, and object recognition (Ullman et al., 2002). According to a traditional view of visual perception, it involves low-level and high-level processing stages (Lamberts & Goldstone, 2005). The early visual area encodes raw information in the low-level processing stage, and the task is to interpret this raw information to enable the perception of people or objects in reality (i.e., object recognition) in the high-level processing stage (Ullman, 1996). Generally, the human visual system efficiently extracts local elements from cluttered backgrounds and integrates these elements into meaningful perception, which is a critical step before object recognition (Li et al., 2017).

Given this context, we hypothesize that contact-analog warnings, which provide the contour of unexpected stimuli compared to bounding box warnings, involve higher levels of visual processing. Bounding box warnings direct the driver's attention to the hazard through a simple, salient geometric shape, leading to a generalized representation of the visual warning in a short period (e.g., the spatial presence of the imminent hazard). This process involves low-level visual processing. In contrast, contact-analog warnings provide the complex contour characteristics of critical hazards. Contours often play an essential role in defining the types of objects to be recognized (Li et al., 2017). Drivers not only sense the spatial presence of the visual warning but also interpret and recognize the contour of the contact-analog warning (i.e., judging whether it is a pedestrian, a vehicle, or other driving-related objects). This process involves high-level visual processing. Information theory suggests that the more information a graphic has, the more effort is required to process it (Rigau et al., 2005).

Consequently, there are variations in the time required to process responses to these warnings due to differences in visual processing levels. For instance, drivers only need to recognize the simple shape of the bounding box and localize potential hazards by the location where the bounding box warnings appear, resulting in a shorter response time. Conversely, drivers have longer response times for the contact-analog warning because they not only locate potential hazards through the location of the contact-analog warning but also recognize the contour of the contact-analog warning to determine the type of hazard. This result is consistent with the findings of Winkler et al. (2018), who investigated the effect of the degree of specificity of warnings. The results showed that generic warnings allow for better results by triggering a driver response even when the type of safety-critical object has not been identified. They proposed that differentiating the warning might add to drivers' already high cognitive load. Given the dynamic nature of AR graphics, the driver not only needs to interact with the dynamic environment but also recognize the warning which affected by the environment (Kim et al., 2016). Therefore, the driver's perceptual processing process may be more difficult. It is crucial to provide warnings in a form that corresponds to the drivers' perceptual ability.

Additionally, a notable difference in response time was observed within the peripheral vision for these two AR warnings. Considering the limited capacity of the peripheral vision for processing detailed information (Vater et al., 2022), these findings suggest that graphics requiring high-level processing stages may not be suitable for presentation in the peripheral vision. Moreover, the peripheral field of view primarily serves to detect the scene gist and direct the driver's attention to it (Larson & Loschky, 2009; Vater et al., 2022). Currently, due to technical limitations, the existing ADASs primarily use head-down displays (HDDs), while common head-up displays (HUDs) have limited presentation areas, mostly within 3–5 inches (Van Krevelen & Poelman, 2010). Consequently, warning information is predominantly presented in the driver's peripheral vision. Investigating the

perceptual capabilities of graphics that the peripheral vision can effectively process is crucial, as is striking a reasonable balance between information provision and the processing level of graphics. This finding may also encourage future research on road signs to consider appropriate simplification of their graphic presentations, especially considering that road signs and other signage are primarily located in the peripheral field of view (Costa et al., 2018).

In summary, previous studies have already explored methods for adjusting the physical parameters of the HMI interface to improve the presentation of warning-related information (e.g., Morita et al., 2007; Wang et al., 2022). However, an intriguing aspect of warning design is considering the perceptual level of the warning. When it comes to auditory warnings, drivers face challenges in processing and interpreting the information conveyed through auditory icons and earcons (Amer & Johnson, 2018). The findings of this experiment suggest that a similar phenomenon may also apply to visual graphical warnings. Subsequent research should address how to strike a balance.

5. Limitations

There are several limitations to consider when interpreting the findings of this study. Firstly, it should be noted that video clips were used as materials in accordance with previous studies (e.g., Chen et al., 2023; Eyraud et al., 2015; Hwang et al., 2016; Wang et al., 2021). As a result, the reaction tasks employed in this study, such as watching video clips and pressing buttons in response to unpredictable events, were relatively simple and lacked complexity. Therefore, it is uncertain whether these findings can be generalized to more complex driving conditions. Secondly, despite the use of real AR devices to provide a realistic AR driving experience, issues related to depth perception and focal switching distance still remain due to the overlay of virtual images on physical objects. These factors could potentially affect the validity of the results. Thirdly, it is important to acknowledge that the real traffic environment is significantly more complex than the simulated scenarios used in this study. This raises concerns about the ecological validity of the findings, as the actual driving conditions may introduce additional variables not accounted for in the experimental design. Fourth, in this study, we are concerned with visible hazards, while the difference in effectiveness between the contact-analog warning and the bounding box warning for invisible hazards remains to be investigated. Furthermore, it is important to note that the between-subjects design used in this study may result in limited data being collected in each AR warning group. The rationale behind the between-subjects design choice was to maintain a certain level of novelty for participants in response to unexpected stimuli and their AR-HUD warnings, as well as to minimize subject fatigue by keeping the experiment duration relatively short. However, future studies could consider increasing the sample size or utilizing a within-subjects design to further investigate the effectiveness of AR-HUD warnings. Lastly, it should be noted that the majority of

participants in this experiment were Chinese novice drivers without prior experience in using AR-HUD devices. Considering that drivers' familiarity with the system may influence their performance (Dingus et al., 1997), future studies should aim to recruit participants with more driving experience or prior experience using AR-HUD devices to further explore the effects of AR warnings. Given these limitations, additional research is necessary to validate and broaden the current findings, to achieve a more comprehensive comprehension of the influence and efficacy of AR-HUD warnings in diverse driving circumstances and among various driver demographics.

6. Conclusion

The objective of this study was to examine the impact of two types of AR warnings, namely the contact-analog warning and the bounding box warning, in different collision scenarios and traffic densities. Our findings indicate that both types of AR warnings were effective in both high and low-traffic-density conditions. Furthermore, we recommend incorporating different AR warnings into the design of the AR interface based on the specific collision types. In the case of rear-end collisions, both the bounding box warning and the contact-analog warning were effective. However, the contact-analog warning demonstrated slightly higher suitability for rear-end collisions. Conversely, for pedestrian collisions, the bounding box warning was found to be more effective compared to the contact-analog warning. These findings highlight the importance of tailoring AR warnings to specific collision types. By adhering to these recommendations, future AR interface designs can optimize the use of different AR warnings to enhance driver safety in various collision scenarios and traffic conditions. Furthermore, we analyze the aforementioned two types of graphs in connection with the cognitive processing perspective. The findings also suggest that, in the future, the field of warning design should consider the potential impact on warning effectiveness arising from the varying perceptual processing levels of the graphs in individuals.

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